Planta Física: four decisions

Challenge and assessment guide · MA2015.202

Version: October 6, 2026

1. We practice the fundamentals

We use the supplied functional genetic algorithm to analyze treated-water allocation to four buildings for one month. Keep one cumulative notebook with formulation, calculations, code, results and interpretation. Research and Final can receive full credit with the supplied base when you demonstrate understanding and meet their criteria. Development requires one substantive operator change.

We work with Planta Física at the Mexico City Campus and aggregate March 2025 demands. We distinguish observed consumption from teaching assumptions about capacity and limits: results describe an academic model, not a validated hydraulic or sanitary solution.

2. We define four genes and one capacity constraint

Building	Demand D_i (m^3)	Gene k_i	Fraction x_i
Aulas 1	847.00	$0, \ldots, 8$	$k_1/10$
Biblioteca	462.00	$0, \ldots, 8$	$k_2/10$
Cedetec	284.21	$0, \ldots, 6$	$k_3/10$
Aulas 2	683.00	$0, \ldots, 8$	$k_4/10$

Retain the table order. We define treated water T, potable water P and excess E:

$$T = \sum_{i=1}^{4} D_i k_i/10, \qquad P = 2276.21 - T, \qquad E = \max(0, T - 1410).$$

We minimize P subject to the domains and $T \leq C$, with $C = 1410$ m^3 as a **teaching assumption**. We rank candidates using scalar fitness, where lower is better:

$$F(k) = \begin{cases} P/2276.21, & T \leq C, \\ 1 + E/C, & T > C. \end{cases}$$

We use a numerical tolerance of 10^{-8} m^3. We reject invalid encodings without silent rounding or repair. Operators preserve gene domains; the evaluator checks capacity.

Reference calculation. For $(6, 6, 6, 6)$ we obtain $T = 1365.726$, $P = 910.484$ m^3 and $F = 0.4$: it is feasible. For $(8, 8, 6, 8)$ we obtain $T = 1764.126$ and $E = 354.126$ m^3: gene domains are valid, but capacity is exceeded.

Provenance. We use the four March 2025 records in `consumo_agua_2023_2026.csv`, without imputation, zero or anomaly flags. We take the capacity from `parametros_reto.csv`, using the instructor calibration. The notebook metadata retains source records and hashes. Building limits are also teaching assumptions.

3. Build and submit incremental work

The supplied base includes a shared notebook, embedded data and `pyproject.toml` and `uv.lock`. Prepare the environment with `uv sync --locked` and open Jupyter with `uv run jupyter lab`. Reuse these files, add your team explanation and identify any team changes. Retain prior case analysis that remains relevant.

The base uses tournament selection of two, one-point crossover, discrete per-gene mutation, generational replacement and one elite. In Development, modify one operator substantively and compare base versus variant with 20 individuals, 20 generations, seed 42 and $p_m = 0.25$ in both runs. In Final, compare $p_m = 0.25$ and $p_m = 0.50$ using the same identified code and all other conditions fixed. If you still have only the base, identify it and complete the Final analysis with that code; the Development penalty will not be applied again.

Milestone and weight	New evidence in the same notebook	Due date
Research · 15%	Formulation, data and assumptions; one candidate calculated by hand and checked (A1/A2: 60/40).	13/10/2026 23:59
Development · 15%	Substantive operator modification and controlled base/variant comparison (B1/B2: 70/30).	16/10/2026 23:59
Final submission (Innovation) · 30%	Generation explanation, same-code comparison and physical interpretation (C1/C2/C3: 40/35/25).	20/10/2026 16:10

At each deadline, submit the cumulative notebook and the two supplied environment files. Incorporate corrections as the basis for the next milestone. New evidence will be assessed against its own criteria, and the grade already assigned to the previous milestone will be retained.

The final submission contains the completed work. Do not add a separate report, interface, plan exports or multiple-seed experiment. A numerical improvement is not required for a grade of 100.

4. Organize four construction sessions

Session	Construction and closure
Session 5 (October 6)	Define genes, model and evaluator; check calculations and orient Research, due October 13 at 23:59.
Session 6 (October 9)	Trace selection, crossover and mutation in the supplied base.
Session 7 (October 13)	Compare the base with a substantive one-operator variant. Development is due October 16 at 23:59.
Session 8 (October 16)	Compare mutation probabilities in the same identified code, interpret results and reproduce the notebook.

Rotate programming, review and verification responsibilities within your team. Record this rotation in a short notebook cell. If your team has four members, assign comparison interpretation as a separate responsibility. Collaboration evidence accompanies the work without an additional submission or additional points.

5. How the rubric will be applied

Each assignment will be graded out of 100. An evidence-based level will be assigned to each criterion: **I = 40%, II = 70%, III = 90%, IV = 100%**. A score of 0 will be assigned for absent evidence. Each criterion maximum will be multiplied by its level factor, and the results will be added. Executing a cell alone does not establish understanding.

Research · 100 points

A1 · 60 · Formulation and assumptions

I You identify the case but confuse genes, units or constraints.
II You define four genes and the objective; material errors remain in domains, capacity or provenance.
III You give a correct formulation with a minor omission in units, order or the data/assumption distinction.
IV You define order, domains, units, objective and capacity; you distinguish observed demands from assumptions and explain each gene.

A2 · 40 · Hand calculation and check

I You show a partial calculation that cannot verify a candidate.
II You calculate volumes but make a material error in feasibility or fitness.
III You correctly calculate and check a candidate, with an incomplete explanation of a difference or constraint.
IV You show the arithmetic for one candidate, check it against the evaluator and explain why the maximum plan is infeasible despite valid domains.

Development · 100 points

B1 · 70 · Functional cycle and justified modification

I You show isolated operators or a cycle that does not produce a usable next population.
II You run the base without the required substantive modification, or implement a change with a material failure in selection, crossover, mutation, replacement or elitism.
III You run the complete variant cycle with a fixed population, valid genes and a protected elite; you identify a relevant change but leave a minor omission in its rationale or a step's explanation.
IV You run the complete variant cycle with a fixed population, valid domains and the best individual protected; you identify each step and justify a substantive change from an explicit limitation and hypothesis.

If only the base is run, B1 will be capped at level II. Changing only an excluded parameter or code presentation will count as no required modification.

B2 · 30 · Reproduction and controlled comparison

I You show outputs without an identifiable configuration or complete execution.
II You reproduce only one version, depend on undocumented prior steps or state, mix comparison conditions, or fail to connect results to genes, volumes and feasibility.
III You reproduce both versions from beginning to end with explicit environment, seed and shared conditions; you compare results correctly, with a minor omission in configuration or interpretation.
IV You run and reproduce both versions from the beginning with explicit environment and seed; you present a controlled comparison, interpret genes, volumes, fitness and feasibility, and explain whether the evidence supports the hypothesis and the limit of one seed.

If only one executed version is presented, B2 will be capped at level II. Repeating the base and naming it a *variant* without an operator change still counts as one version.

Final submission (Innovation) · 100 points

The explanation and analysis added to the cumulative notebook will be assessed. Development code will be used as supporting evidence; its implementation criterion will not be assessed again. The base without modification is an eligible analyzed version. Development caps and any penalty for not modifying the base will not be applied again in Final.

C1 · 40 · Explanation of one generation

I You name steps without explaining how the population changes.
II You describe the sequence with material confusion between selection, crossover, mutation and replacement.
III You explain the sequence correctly, with a minor omission about elitism or fitness direction.
IV You trace concrete candidates from parents to replacement; you explain each operator, why lower fitness is better and how the elite is protected.

C2 · 35 · Comparison of two configurations

I You show two outputs without identifying conditions or comparing results.
II You identify the change but mix other conditions or misinterpret the plot.
III You compare controlled conditions with a correct plot, but incompletely explain equal results, differences or plateaus.
IV You change only mutation, show configuration and plot, compare results and explain limits; a tie or a worse result is also valid.

C3 · 25 · Physical interpretation and coherence

I You present numbers without linking them to buildings, units or capacity.
II You partly connect results to the case but omit a constraint or claim unsupported operational validity.
III You correctly interpret allocation, volumes and capacity, with an incompletely explained model limitation.
IV You explain building fractions, potable and treated water, capacity and assumptions; your claims match outputs and respect the monthly teaching scope.

6. Written team code examination · 10%

Answer, with your team, written questions prepared from your submitted notebook and code. The assessment will be administered on 23/10/2026 from 17:00 to 17:50, after the individual article examination. Questions will be based on chromosomes and fitness, one operator, and results or reproduction. Submit one response set as a team; the same grade out of 100 will be assigned to every team member. No oral review will be administered.

One holistic rating will be assigned:

IV · 100
: You answer all three areas accurately using your code and results, justify decisions and solve a simple variation of the case.

III · 90
: You answer all three areas correctly with a minor omission or brief clarification.

II · 70
: You demonstrate partial understanding, require substantial help in one area or confuse a central decision.

I · 40
: You recognize parts of the code but fail to connect the model, operators and result.

No evidence · 0
: You submit no assessable responses.

Identify the supplied base and your team contributions in your written responses. Understanding of the submitted code and results will be assessed; points will not be assigned for the number of extra features.

7. How the course grade will be calculated

The individual written article examination retains its existing instrument and 30% weight; take it on 23/10/2026 from 16:10 to 17:00. The course grade will be calculated from five grades out of 100:

$$N = 0.15A + 0.15B + 0.30C + 0.10K + 0.30E.$$

Each weight will be applied once. For example, a Research grade of $60(1) + 40(0.70) = 88/100$ will be assigned when A1 is at IV and A2 is at II; $0.15(88) = 13.2$ points will be included in the course grade.

8. Check your submission

- Run the entire notebook after restarting its kernel and retain outputs.
- Identify data, assumptions and configuration, and check your team hand calculation.
- Explain one generation with concrete candidates and connect operators to your team code.
- Include the two required configurations and a readable plot; interpret ties or differences without requiring improvement.
- Retain conclusions and team rotation in the notebook alongside the supplied environment files.

The extraordinary partner presentation, if scheduled, will be kept separate from assessment; no additional points will be assigned.